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Reflection Journal

Working through this lab changed the way I understand image processing and its connection to modern AI tools. Before completing the exercises, it was easy to think of AI image systems as “black boxes” that magically enhance, classify, or transform images. But implementing the operations manually—convolution, edge detection, histogram equalization, geometric transforms, and multi-step creative effects—made it clear that these classical techniques are the mathematical backbone of today’s AI systems. The difference is scale: in traditional processing, I apply the operations directly, while AI models learn and combine them automatically.

Seeing this connection helped me appreciate how much of AI is built on simple, interpretable components. For example, the convolution filters I applied manually are the same type of filters that convolutional neural networks learn in their early layers. The geometric transforms I used for resizing and rotation are the same operations used in data-augmentation pipelines. Even the creative effects I built—blending color shifts, blurs, and edges—mirror the layered transformations used in style-transfer and generative models. Understanding the math behind these operations makes AI feel less mysterious and more like an extension of the tools I already know.

I also gained a clearer sense of the trade-offs between traditional and AI-powered approaches. Traditional methods are transparent, predictable, and computationally efficient. I always know exactly what a Sobel filter or histogram equalization is doing. AI methods, on the other hand, are far more flexible and powerful, but they can be harder to interpret and require significantly more data and computation. Traditional techniques give control; AI gives adaptability. Both have strengths, and both matter.

The real-world applications of these techniques are everywhere. Medical imaging relies on contrast enhancement and edge detection to help radiologists interpret MRIs, CT scans, X-rays, and ultrasounds. Satellite imaging uses geometric transforms, histogram equalization, and noise reduction to analyze weather patterns and environmental changes. Robotics depends on image processing for object detection and navigation. Self-driving vehicles use these same operations for lane detection, obstacle recognition, and traffic-sign reading. Seeing these connections made the lab feel practical and relevant.

One of the most interesting parts of the lab was experimenting with creative effects. The vintage effect stood out because it combined multiple operations—color-channel adjustments, contrast reduction, Gaussian blur, and tone shifting—to create a cohesive artistic style. It demonstrated how simple operations, when layered intentionally, can produce complex and visually compelling results. It also reinforced how AI models build on these same principles, just with learned parameters instead of hand-crafted ones.

Personal Experiments Beyond the Basic Requirements

To deepen my understanding, I completed three personal experiments that explored how different combinations and sequences of operations affect the final output. These experiments helped me see how classical techniques behave in practice and how small changes can produce dramatically different results.

Experiment 1: Blur Before vs. After Edge Detection

In this experiment, I tested how the order of operations affects edge detection. I created two versions of the same image:

- Version A: Gaussian blur → Canny edge detection
- Version B: Canny edge detection → Gaussian blur

Blurring first produced cleaner, more stable edges, because noise was reduced before the edge detector ran. Blurring afterward created a soft, sketch-like effect, showing that even with the same operations, the sequence matters. To be honest, the image makes my cat look like something out of a horror movie! This experiment demonstrated how pipeline design directly influences the final output, a principle that carries over into AI systems where layer order determines learned behavior.

Experiment 2: Contrast Stretching vs. Histogram Equalization

Next, I compared two different contrast-enhancement techniques on the same grayscale image:

- Contrast Stretching: A simple linear rescaling of pixel intensities
- Histogram Equalization: A redistribution of intensities to enhance detail

Contrast stretching produced a subtle improvement, making the image slightly clearer. Histogram equalization created a much more dramatic enhancement, revealing details in shadows and highlights that were not visible before. This experiment helped me understand why histogram-based methods are widely used in fields like medical imaging, satellite analysis, and forensic enhancement—anywhere detail recovery is essential.

Experiment 3: Vintage vs. High-Clarity Style Pipelines

For my final experiment, I built two different creative styles using combinations of basic operations:

- Vintage Effect:
 - Gaussian blur
 - Reduced contrast
 - Warm color shift (increasing the red channel)
- High-Clarity Effect:
 - Sharpening filter
 - Increased contrast
 - Slight saturation boost

The vintage effect produced a soft, nostalgic look, while the high-clarity effect created a sharp, crisp, high-contrast image. This experiment showed how simple operations, when combined intentionally, can create distinct artistic styles. It also mirrored how modern AI image-generation models layer transformations to produce stylistic outputs.

Connecting Classical Techniques to AI

Completing these experiments made the connection between classical image processing and AI-based systems much clearer. Traditional methods are:

- Transparent — I know exactly what each operation does
- Predictable — the results follow directly from the math
- Efficient — they run quickly and require no training

AI methods, on the other hand, are:

- Flexible — they learn patterns automatically
- Powerful — they can handle complex tasks like recognition and generation
- Less interpretable — the internal transformations are not always obvious

Both approaches matter. Classical techniques give control and understanding; AI techniques give adaptability and scale. This lab helped me see how they complement each other rather than compete.

Looking Ahead

Looking ahead to my Adventure Quest, I want to explore how traditional image processing connects to modern AI-based techniques in real investigative and forensic contexts. I'm especially interested in how organizations like the FBI use tools such as edge detection, sharpening, and enhancement to analyze images, read license plates, and interpret locations. I want to understand how the classical operations we used in the lab became the foundation for today's advanced forensic systems, and how older methods were incorporated into newer AI-driven approaches.

Finally, if I were to combine traditional and modern techniques in a project, I would build a system that analyzes vehicle traffic patterns in my neighborhood. Traditional image processing could detect and track vehicles, while AI models could classify patterns such as speeding trends, congestion, or peak traffic times. This would allow me to use classical operations for precise measurement and AI for higher-level interpretation—a balanced, practical blend of both worlds.

Conclusion

This lab gave me a deeper, more grounded understanding of how image processing works and how it connects to modern AI. By completing the required tasks and conducting three additional experiments, I gained hands-on experience with how operation order, contrast methods, and stylistic combinations influence results. More importantly, I learned how classical techniques form the backbone of today's AI systems. This experience made the entire field feel more accessible, logical, and interconnected.